Rimsha Sarfaraz

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Education

Saarland University — Saarbrücken, Germany
Masters of Science in Computer Science — October 2025 — July 2027

Lahore University of Management Sciences (LUMS) — Lahore, Pakistan
Bachelor of Science in Computer Science — August 2020 — May 2024

Publications

Sumair Ijaz Hashmi*, **Rimsha Sarfaraz***, Lea Gröber, Mobin Javed, Katharina Krombholz. Understanding the Security Advice Mechanisms of Low Socioeconomic Pakistanis *2025 CHI Conference on Human Factors in Computing Systems.*
*(Best Paper Honorable Mention; *joint first authorship)*

Research Experience

CISPA Helmholtz Center for Information Security - *Graduate Research Assistant* — October 2025 — Present
Empirical & Behavioral Security — *Advisors: Dr. Katharina Krombholz & Dr. Lea Gröber (ICSI, UC Berkely)*

Mobile Phone Shopkeepers as Intermediaries for Low Socio-economic Pakistanis

- Crafted a qualitative interview guideline for investigating low-end mobile phone users' reliance on shopkeepers for phone/app setup and their security practices by conducting a thorough literature review and pretests.
- Currently performing a thematic analysis and preparing the results *to submit to USENIX 2027 as a co-first author*.

Usability of Google Account Creation Process for Low-Income Pakistani Women

- Designed an extensive research protocol for a randomized controlled trial (RCT) to evaluate the usability of the Google account onboarding process among women from low-socioeconomic backgrounds. Conducted pretests to evaluate and select accessibility tools.
- Developed structured interview guidelines to standardize data collection across researchers, who acted as intermediaries while participants completed Google account creation tasks. Collaborated with a data collection team to pilot and iteratively refine the research protocol. *Results to be submitted to USENIX 2027*

Non-commercial Visual Artists' Perceptions of Machine Unlearning & Relevant AI Policy

- Prototyped a platform to host a Machine Unlearning model that attempts to unlearn a selected artist's style.
- Researched AI copyright-related policies across regions and conducted pre-tests to develop and improve an interview guide for conducting a demo of the Machine Unlearning model and discussing AI policies with non-commercial visual artists. *Results submitted to ACM CHI 2027*

Lahore University of Management Sciences (LUMS) - *Research Assistant* — January 2023 — September 2025
Security & Analytics Lab — *Advisors: Dr. Lea Gröber, Dr. Mobin Javed & Dr. Katharina Krombholz*

Security Advice Mechanisms of Low Socioeconomic Pakistanis

- Co-led a qualitative study on how low-income, low-literate Pakistanis learn and share security advice to adopt secure behavior. Focused on what advice they learn, how do they learn it, and why (or why not) do they share it with other people.
- Crafted a context-sensitive interview protocol to understand the security mental models, security advice, and advice sharing practices of low-income, low-literate Pakistanis
- Conducted semi-structured interviews with women Janitorial staff at a University and factory workers in a textile factory.
- Published an *award wining* academic conference paper at *ACM CHI 2025*

Internet, Data & Society Lab — *Advisors: Dr. Ihsan Ayyub Qazi & Dr. Ayesha Ali*

Game-based Intervention to Counter Fake News

- Designed and developed an interactive story-based game to improve truth discernment and promote digital media literacy.
- Researched psychological interventions (e.g., inoculation theory, accuracy nudges) and game design principles to inform gameplay mechanics. Conducted Usability testing to improve the design iteratively.

Tailored Digital Media Video Interventions To Counter Fake News

- Designed four culturally relevant video interventions to teach low-digital-literacy audiences how to identify fake news and deepfakes.
- Curated datasets for baseline and end-line surveys, including producing audio-deepfake videos and extracting true/fake audio samples.
- Conducted qualitative analysis of participant responses to identify reasoning patterns for categorising news as true or fake.

Interactive Media Lab (Awaaz-e-Sehat) *Advisor: Dr. Maryam Mustafa*

Empowering Maternal Health Through AI

- Led user experience research and interaction design for an AI-enabled maternal health application. Investigated how AI could support healthcare practitioners' workflows and improve prenatal care delivery.
- Conducted field observations in a hospital setting, studying doctors' and nurses' workflows that informed the requirements and design of the application.
- Conducted a quantitative error analysis of AI-generated Electronic Medical Records (EMRs) against ground-truth records, measuring error rates, categorizing error types, and assessing their potential implications for clinical decision-making and patient diagnosis.

Teaching Experience

Teaching Assistant, *Humans of Computing: Past, Present & Future (Seminar)*, Saarland University — October 2026 — Present

Teaching Assistant, *Introduction to Data Science (Machine Learning Specialization)*, LUMSx — September 2024 — April 2025

Summer School Instructor, *Navigating AI Tools*, Summer School @ LUMS — Summer 2024 & Summer 2025

Teaching Assistant, *Building Research Skills*, Winter School @ LUMS — December 2024 — January 2025

Teaching Assistant, *Data Science for Health Care Professionals*, LUMS Learning Institute — February 2024 — March 2024

Talks

Paper Presentation ACM CHI 2025 (Link)

Honors & Awards

ACM CHI Best Paper Honorable Mention Award — 2025

SIGCHI Gary Marsden Travel Award — 2025

LUMS Students as Co-Researchers Grant — 2023

Community

Student Volunteer — April 2025
ACM CHI 2025

Founding Member & Vice President — November 2020 — June 2024
LUMS Women in Computing

Organizer — January 2024 — February 2024
Google exploreCSR @ LUMS

Lead Python Instructor — February 2024
Coding bootcamp - LUMS Women in Computing